Sahaj Bhadja

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Summary

Data-focused Software Engineer with 2+ years in **Python**, **SQL**, and FastAPI, building data pipelines, **machine learning** models, and spatial data workflows, with strong data-quality validation, reporting, and clear communication of findings.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **FastAPI** services and ETL-style pipelines extracting and transforming data across PostgreSQL, SQL Server, and cloud storage with incremental loads and **data validation**, processing 5K+ monthly jobs at 99%+ reliability.
- Designed normalized multi-tenant **SQL** schemas for 200+ users and 10K+ records, applying indexing, reconciliation checks, and query tuning that reduced latency 35% and improved **data quality** across integration workflows.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python pipelines over 100K+ trajectory logs and HD-map **vector geometry**, including road boundaries and lane connectivity, aligned to a global coordinate frame to train a PyTorch model that improved planning 18%.
- Designed a **spatial-temporal encoder** with distance-aware attention and **validated** it against three baselines in simulation with Weights & Biases tracking, improving planning stability 15% and documenting failure modes.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL** on AWS Glue and S3 to transform raw JSON, CSV, and Parquet from multiple sources into normalized analysis-ready datasets, improving pipeline efficiency 68% and cutting storage costs 15%.
- Consolidated user activity logs into unified formats for analytics, wrote 20+ PyTest and Mockito **automated tests**, and optimized API queries with indexing and caching to improve response times 30%.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Engineered **21 behavioral features**, including IP presence, URL length, and suspicious keywords, over a 650K+ URL dataset and trained Gaussian Naive Bayes and **Random Forest** classifiers for phishing and malware detection.
- Compared classical models against RNN, LSTM, Bi-LSTM, GRU, and CNN architectures, achieving **96%+ detection accuracy** and generating actionable insights into phishing and malware trends across the dataset.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built hourly **Spark** micro-batch pipelines on Databricks over 50K+ booking and payment events, aggregating peak hours, most-used services, daily active users, and revenue into **dashboards** for stakeholders.
- Presented the end-to-end data and system architecture to the client before implementation, then built Flask APIs and pricing and coupon logic that kept core booking responses under 200ms.

Skills

Languages & Databases: Python, SQL, Java, C++, Bash, PostgreSQL, SQL Server, MySQL
Data Science & ML: Feature Engineering, Random Forest, Naive Bayes, Hyperparameter Tuning, PyTorch, TensorFlow
Data Engineering: ETL Pipelines, Spark, Databricks, AWS Glue, Parquet, Data Validation, Data Modeling, Query Optimization
Spatial & Experimentation: Coordinate Transforms, HD-Map Vectors, Spatial Features, Tableau, Dashboards, Weights & Biases
APIs & Tools: FastAPI, Flask, REST APIs, AWS, Docker, Git, GitHub Actions, PyTest, Streamlit, Gradio, Transformers, Hydra

Projects

Trade Area Site Scoring Model | *Python, GeoPandas, PostGIS, scikit-learn, FastAPI, ArcGIS (arcpy)*

- Designed a **geospatial** model that builds drive-time trade areas, joins census demographics and competitor point-of-interest density, and ranks candidate sites with gradient boosting under spatial cross-validation, served through **FastAPI**.

Store Network Expansion Optimizer | *Python, OR-Tools, GeoPandas, H3, matplotlib*

- Designed an **optimization** model on an H3 hexagon demand grid that selects new locations maximizing incremental delivery coverage while penalizing cannibalization of existing stores, with matplotlib **maps** comparing scenarios.

CaptionCraft Live | *Transformers, PyTorch, Flask, Streamlit, Gradio*

- Benchmarked BLIP, GIT, and ViT-GPT-2 transformer models on a 10K-image, 100K-caption dataset, **validating** fine-tuned variants that raised caption accuracy from 81% to 96% and serving them through inference APIs.